

Efficient Pyramid-Structured Networks Using Channel Separating Gate

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Abstract—Convolutional neural networks (CNNs) have been brought noticeable improvement on numerous image processing applications. In particular, spatial pyramid structures have been widely used, but the number of required parameters increases exponentially as the network deepens. Therefore, we propose a channel separating gate (CSG) that can determine the number of channels in convolutional layers according to the spatial level. For a given spatial level, the CSG separates the channels of feature maps for use at the smaller level. Consequently, the standard non-pooling model can be altered to an efficient pyramid structure by performing the proposed method repeatedly. For single image super-resolution (SR) and deblurring, we demonstrate that our method improves the baseline network in terms of peak signal-to-noise ratio (PSNR) and floating-point operations (FLOPs).

Index Terms—convolutional neural network (CNN), pyramid structure, super-resolution (SR), deblurring

I. INTRODUCTION

In the last few years, convolutional neural networks (CNNs) have outperformed the conventional methods in many computer vision tasks. Despite the improvement of computing hardware, however, it is still challenging to apply the CNNs on mobile devices. Recently, active research has been conducted to achieve optimal performance within limited resources. Since convolutional layers in the CNNs consider spatial information, it is necessary to design a structure that can aggregate them. Previous researches [1], [2] proposed spatial pyramid structures that utilize feature maps with diverse spatial sizes. Although these structures show promising results, they have a limitation of requiring huge network parameters.

We propose a novel channel separating gate (CSG) to design the output channels of convolutional layers in accordance with spatial level. In general, in the pyramid structure, the number of channels is set to hold the floating-point operations (FLOPs). Since the appropriate computation cannot be determined intuitively, it can lead to redundant computations. Rather, while reducing the FLOPs, the proposed method can build an efficient architecture that performs non-excessive computations by separating the channels of feature maps for two different spatial sizes. Through extensive experiments for a couple of vision tasks, *i.e.*, single image super-resolution (SR) and deblurring, we show that our method can

further improve the performance of the baseline network in terms of peak signal-to-noise ratio (PSNR) and FLOPs.

II. PROPOSED METHOD

Figure 1 illustrates the framework of the proposed method to design an efficient pyramid-structured network. For each spatial level n , the proposed CSG determines a separating ratio r_n that divides the channel c_n of the input feature maps into $r_n c_n$ and $(1 - r_n)c_n$. The feature map of the $r_n c_n$ channels is employed for the smaller spatial level via convolutional layer of stride 2, while the rest passes through the stacked convolutional layers at the existing level. We adopt the standard non-pooling model, where $r_1 = 0$, as the baseline network. Meanwhile, setting all separating ratios to 1 implies the most commonly used way of the pyramid structure, *i.e.*, U-Net [1]. Specifically, the feature maps of each spatial level are simply concatenated in the decoding stage without convolutional layers.

As shown in Figure 2, in the proposed CSG, the input feature vectors obtained by global average pooling are fed into the multi-layer perceptron. Then, a binarizer following the sigmoid activation function produces the r_n . Consequently, each channel has a value of 0 or 1, which is used for the same or smaller spatial level respectively. To make the layers before the binarizer be trainable, inspired by binarized neural networks (BNNs) [3], [4], we adopted the approximate gradient of 1 during the back-propagation procedure. Since the CSG is not employed during the test, it does not cause additional network parameter increases.

III. EXPERIMENTS

A. Implementation Details

For each spatial level n , we trained the entire network with the CSG determining the r_n in an end-to-end manner. We used a batch size of 16 for 100k iterations and employed the Adam optimizer of β_1 and β_2 were set to 0.9 and 0.999, respectively. We set the initial learning rate to 0.0001 and reduced the learning rate to 1/10 after 50k iterations. Next, for the same iteration scheme, the model established by fixed r_n is trained from random initialization like [5]. During the training, the c_1 was set to 64, and the proposed method was step-by-step applied up to 4 times.

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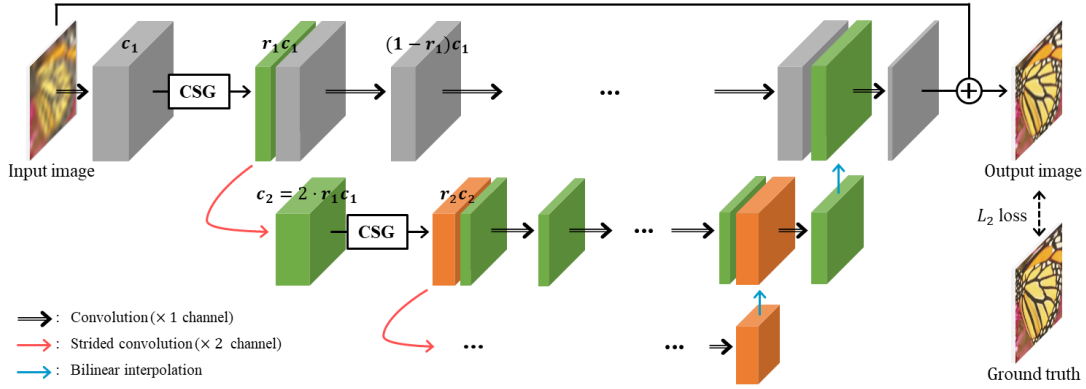


Fig. 1. Application of the proposed method to the standard non-pooling model. The rectified linear unit (ReLU) activation functions are applied after all convolutional layers except the last one.

TABLE I
QUANTITATIVE RESULTS FOR $\times 8$ (TOP) AND $\times 16$ (BOTTOM) SR

Method	Separating ratio		PSNR (dB)					FLOPs	# parameters
	n	r_n	Set5	Set14	BSDS100	Urban100	Average		
Baseline	1	0.00	24.10	22.28	23.05	20.15	21.70	1.000	1.000
Proposed method	1	0.42	24.31	22.44	23.16	20.31	21.84	0.476	1.344
	2	0.42	24.40	22.51	23.21	20.40	21.91	0.306	1.716
	3	0.46	24.35	22.50	23.20	20.36	21.88	0.274	1.782
	4	0.59	24.33	22.48	23.19	20.35	21.87	0.272	1.622
U-Net	1-4	1.00	24.49	22.57	23.26	20.47	21.97	0.703	47.22

Method	Separating ratio		PSNR (dB)					FLOPs	# parameters
	n	r_n	Set5	Set14	BSDS100	Urban100	Average		
Baseline	1	0.00	20.58	19.66	20.97	17.90	19.48	1.000	1.000
Proposed method	1	0.52	20.77	19.80	21.08	18.00	19.58	0.477	1.086
	2	0.50	20.87	19.88	21.15	18.08	19.66	0.383	1.144
	3	0.53	21.18	19.90	21.18	18.11	19.69	0.367	1.116
	4	0.57	20.90	19.89	21.17	18.09	19.67	0.365	1.054
U-Net	1-4	1.00	21.02	20.00	21.27	18.22	19.79	0.703	47.22

TABLE II
DEBLURRING PERFORMANCE EVALUATION ON THE GoPro DATASET

Method	Baseline	Proposed method			
		1	2	3	4
n	r_n	1 0.00	2 0.56	3 0.45	4 0.47
PSNR (dB)		26.33	26.55	26.66	26.62
FLOPs		1.000	0.490	0.412	0.396
# parameters		1.000	0.986	1.112	1.144

For the SR experiments, we trained the network using 800 images from DIV2K dataset and tested it on a number of datasets, *i.e.*, Set5, Set14, BSDS100, and Urban100. Specifically, the high-resolution images were scaled by bicubic interpolation with factors of 8 and 16. For the deblurring tasks, we used 2,053 images for training and 1,111 images for test in GoPro dataset. As input data to be used for training in both experiments, we randomly cropped 96×96 patches and augmented them with rotation and flipping.

B. Experimental Results

We evaluated the effectiveness of the proposed method by measuring the PSNR. At the same time, we presented FLOPs and the number of network parameters to emphasize

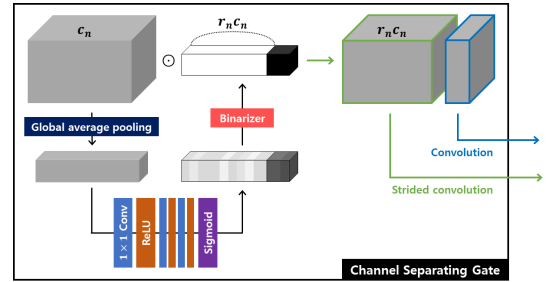


Fig. 2. Separation process of the feature maps in the proposed CSG.

the efficiency of our method. They were normalized based on the baseline network. As shown in Tables I and II, our method not only achieves performance increase on the baseline model but also considerably reduces the FLOPs. In particular, Table I indicates that the proposed method shows comparable performance even with significantly lower network parameters compared to the U-Net. These results demonstrate that the proposed method can attain high efficiency suitable for consumer electronics.

IV. CONCLUSION

In this paper, we introduced a novel channel separating method to design the efficient pyramid-structured networks. The proposed method prevents redundant channels from being used for each space level. The experimental results show that our method can leverage the CNNs-based vision tasks for commercial mobile devices.

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